

Sadeeq Adeyemi

Computer Vision and Perception Engineer

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PROFESSIONAL SUMMARY

Perception engineer focused on multi-object tracking through occlusion for robots and autonomous vehicles. Builds trackers that fuse camera, lidar, radar and event-camera data, and backs every claim with multi-seed ablations and public, reproducible code. Raised identity retention through occlusions from 0% to 89% in simulation, then built the tooling to test the same question on 1,256 real NVIDIA driving clips. Works fluently with AI coding agents.

TECHNICAL SKILLS

Tracking and Estimation: MOT/MOTS, sensor fusion, occlusion reasoning, re-ID, EKF/UKF, particle and PHD filters, JPDA, MHT

Deep Learning and Vision: PyTorch, PyTorch3D, SAM 3, YOLOv8, Vision Transformers, NeRF, segmentation, OpenCV, Open3D

Sensors and Datasets: Camera, lidar, FMCW radar, event cameras; NVIDIA PhysicalAI-AV, KITTI, BDD100K, Argoverse 2

Robotics and Software: ROS 2, Python, C++17, MATLAB, JavaScript, Git, Docker, AWS; HOTA, IDF1, CLEAR-MOT, GOSPA

AI Agents: Agentic development with Claude Code: multi-step coding and research workflows gated by automated tests

RESEARCH AND PROJECTS

Occlusion-Aware Multi-Object Tracking | github.com/sadeeqgandalf/lunar-occlusion-tracking 2026

- Designed a tracker (EKF, Hungarian assignment, Bernoulli existence) that derives detection probability from 3-D line of sight and cast shadows, with a negative-information update for missed detections.
- Raised identities kept through 1–5 s occlusions from 0% to 46%, and to 89% with appearance re-ID (20 seeds, 95% intervals).
- Cut ID switches per run from 75.6 to 6.6 by fusing camera, lidar and a second viewpoint; showed that naive fusion raises them 3.4x.
- Cross-checked against MathWorks GNN, JIPDA and TOMHT; Python and MATLAB ports reproduce the JavaScript engine exactly.
- Rendered camera detections with PyTorch3D; appearance re-ID lifts identities kept after hides over 5 s from 8% to 50%.
- Shipped a live 3-D browser lab backed by 46 automated tests: sadeeqgandalf.github.io/lunar-occlusion-tracking
- Added lidar instance segmentation (0.94 mask IoU) and an event-camera sensor model fitted to real recordings (correlation 0.97).

Occlusion Mining on Real Driving Data | NVIDIA PhysicalAI Autonomous Vehicles dataset 2026

- Mined 10,871 pedestrian occlusion events from 1,256 clips (9.6 million 3-D boxes) with a ray-cast line-of-sight test.
- Implemented fisheye (f -theta) projection and per-point lidar motion compensation; confirmed camera-lidar timing to about 10 ms.
- Built a lidar depth-buffer visibility referee and 3-D review videos showing point clouds, blind zones, tracks and hide timers.

Closed-Loop Autonomous Driving Stack 2026

- Integrated EKF localization, SAM 3 perception lifted to 3-D with lidar and radar, tracking, Frenet planning and LQR/PID control.
- Passed 100 of 100 Monte Carlo runs with no simulator ground truth; with occlusion reasoning off, 5 of 10 runs strike a pedestrian.

Additional Perception Work 2025 – 2026

- Open-vocabulary MOTs with SAM 3 on KITTI and BDD100K (HOTA); lidar moving-object segmentation on Argoverse 2.
- FMCW radar pipeline in MATLAB from raw IF samples to CFAR detections, checked against the CRLB; ViT and NeRF in PyTorch.

EXPERIENCE

Robotics Research Assistant | University of Maine Robotics Laboratory 2025 – Present

- Build mobile robot platforms end to end: sensor integration, wiring, ROS 2 nodes and point-cloud perception pipelines.
- Implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

Machine Learning Engineer and Structural Steel Fabricator | NewTech Steel, Nigeria 2018 – 2023

- Deployed a computer vision model for real-time fatigue and defect detection in steel; automated testing cut test time by 50%.
- Led structural steel fabrication and erection for factory and auditorium projects.

Materials Engineer (CPT) | R.W. Gillespie, Maine: materials testing to ASTM standards 2024

EDUCATION AND CERTIFICATIONS

MS, Data Science and Engineering | University of Maine 2025 – Present

MS, Structural Engineering | University of Maine 2023 – 2025

MSc, Project Management | University of Portsmouth, United Kingdom 2023

BEng, Civil Engineering | Federal University of Technology Akure, Nigeria 2019

Certifications | AWS Certified Machine Learning – Specialty; AWS Certified Solutions Architect – Associate